Course Code:

Lecture - 01

Research Methodology

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COURSE OUTLINE

Course Title	**Research Methodology**

Course Outline:

Purpose of Research; Research Project Conceptualization, Choice of Methods; Elements of a Research Proposal, Operationalization choices and illustrations. Research Design: formulation of research design, pretesting of research instruments and procedures, units of Analysis, time dimension; Experimental design and use of indicators in research, Survey Research: Guidelines for asking question and questionnaires construction, Self-administered questionnaires, Interview and other survey methods; their strength and weaknesses. Sampling: the logic of sampling, concepts and terminologies, population and sampling frames, types of sampling design. Field Studies: Steps in the conducting field study; Evaluation Research: How to carry out evaluation research; Analytical tools in research: qualitative and quantitative methods; Statistical Analyses: Univariate, Bivariate and Multivariate analyses

IMPORTANT INFORMATION – MARKS DIVISON

- Assignments: (Total Marks = 20)
  - Two assignments before Midterm = 5 + 5
  - Two assignments before Final = 5 + 5

- Quiz: (Total Marks = 10)
  - Two quiz before Midterm = 2.5 + 2.5
  - Two quiz before Final term = 2.5 + 2.5

- Midterm Exam = 30 marks

- Final Exam = 40 marks

- Total Marks = 20 + 10 + 30 + 40 = 100

IMPORTANT INFORMATION

Bahria University
Discovering Knowledge

MEMORANDUM

To: All HoDs Info: DGKC, Director Admin, All Principals, DD Exam, ADQA, Manager LDC Memo No: /21/2022 Dated: /0 /11 / 2022	From: Director Academics

FINAL EXAMINATION- SHORT ATTENDANCE/RE-TAKE EXAMINATIONS CASES

References:

A. Monthly Meeting held on 31 October 2022.
B. DG Directive 12/2022 dated 26 July 2022.

1. Apropos reference A, criteria for appearing of students in Final / Re-take Exam having short attendance has been decided during the meeting held on 31 Oct 2022 and appended below:

   a. Following students having short attendance are eligible to appear in Final Term Exam:

      (1) Students having attendance 75% and above during the semester.

      (2) Students admitted in hospital due to medical problem for one week in addition to 25% absence already given in Academic Rules during the current semester.

      (3) Fee defaulters due to financial issues/ scholarships having attendance 75% and above during the semester and attendance record available.

   b. Following students are eligible to appear in Re-Take Examinations:

      (1) Student admitted in hospital due to accident/any medical issue on particular day of Exam or Student met with accident causes immobility.

      (2) Bereavement in the immediate family (Parents, Siblings, grandparents, spouse and children).

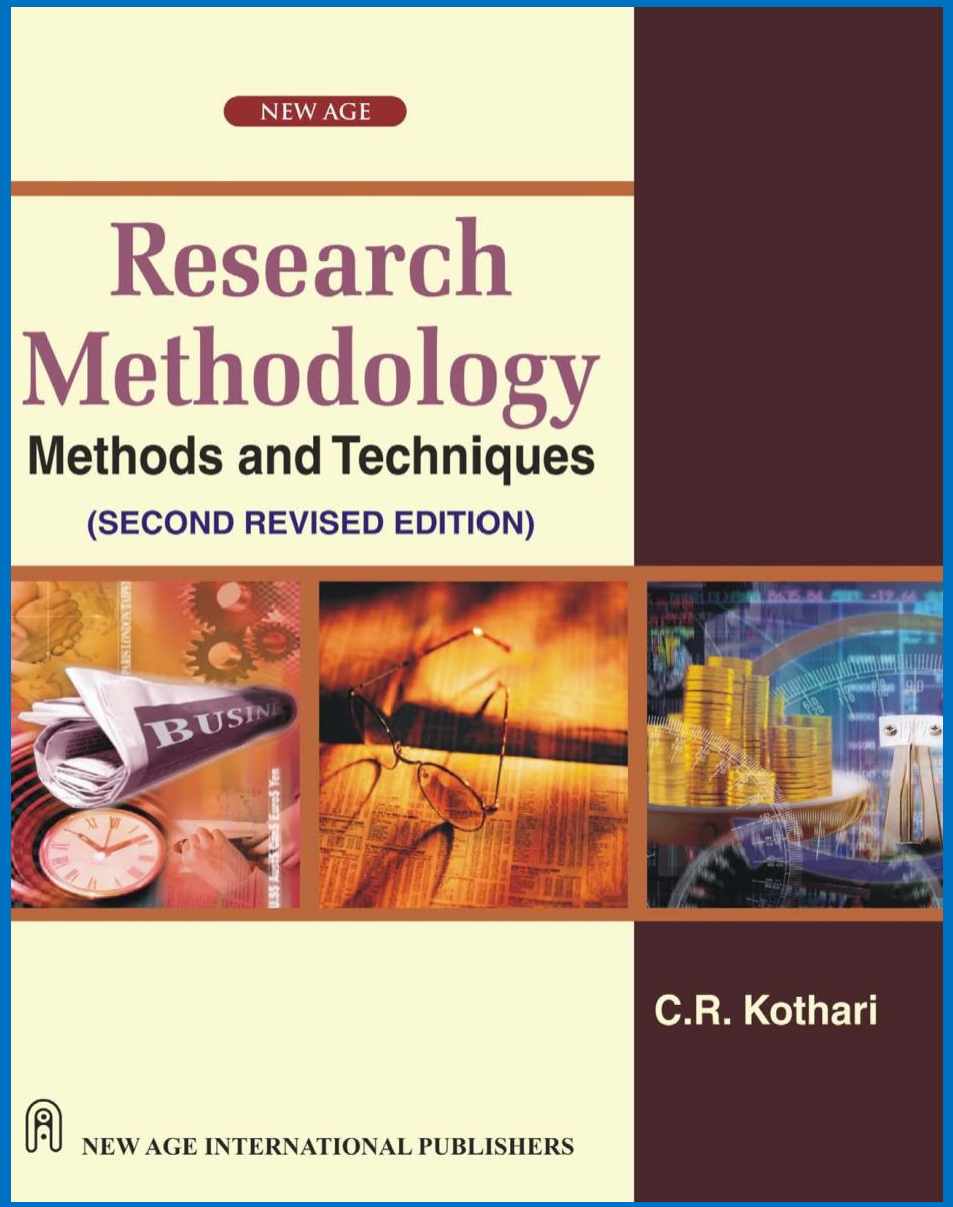
2. *All medical-related cases will be recommended by Medical officer for approval from Director Academics.

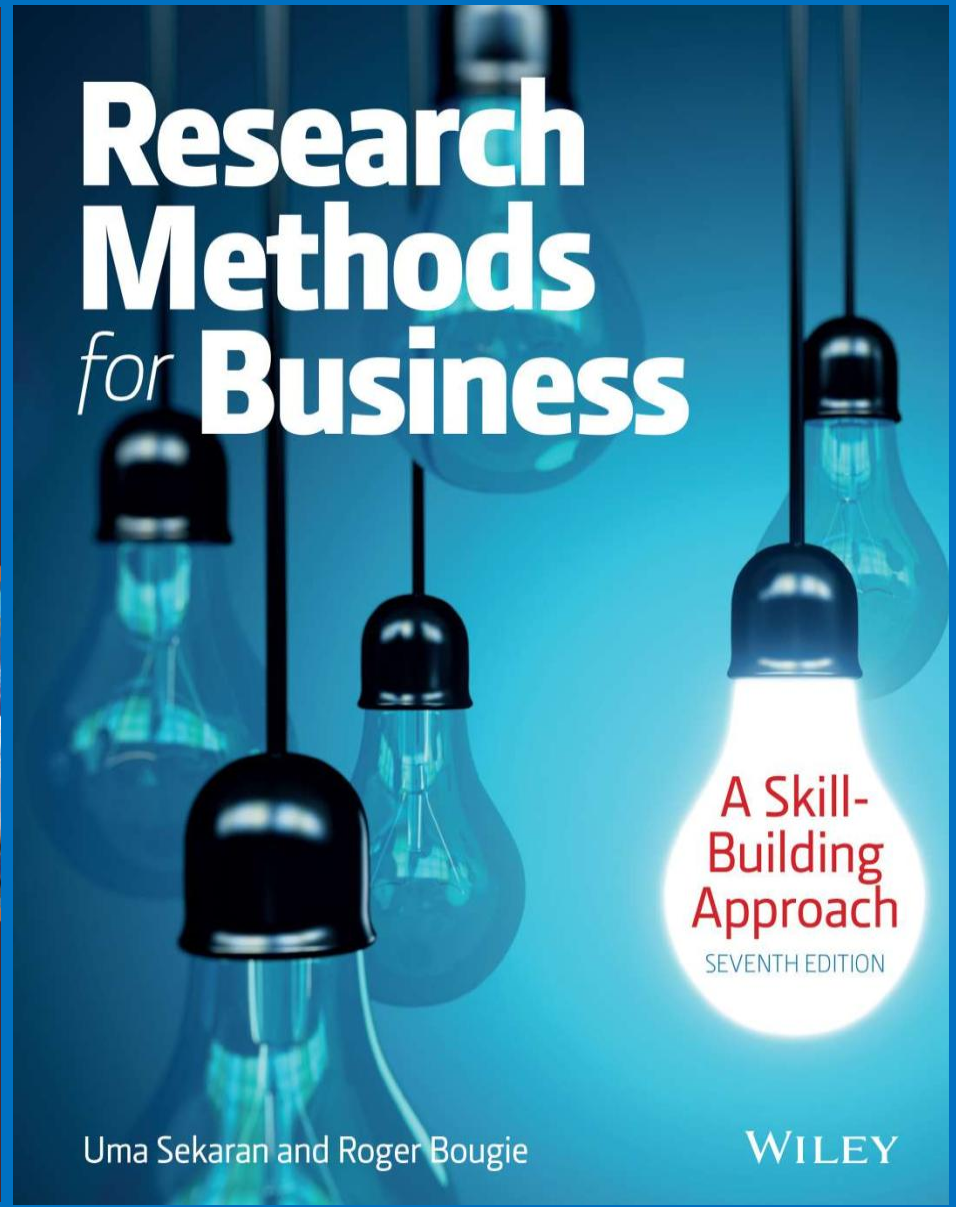
Name: ZAHEER AHMAD Rank: CAPTAIN (R) PAKISTAN NAVY SIGN:

DESIGNATION: DIRECTOR ACADEMICS

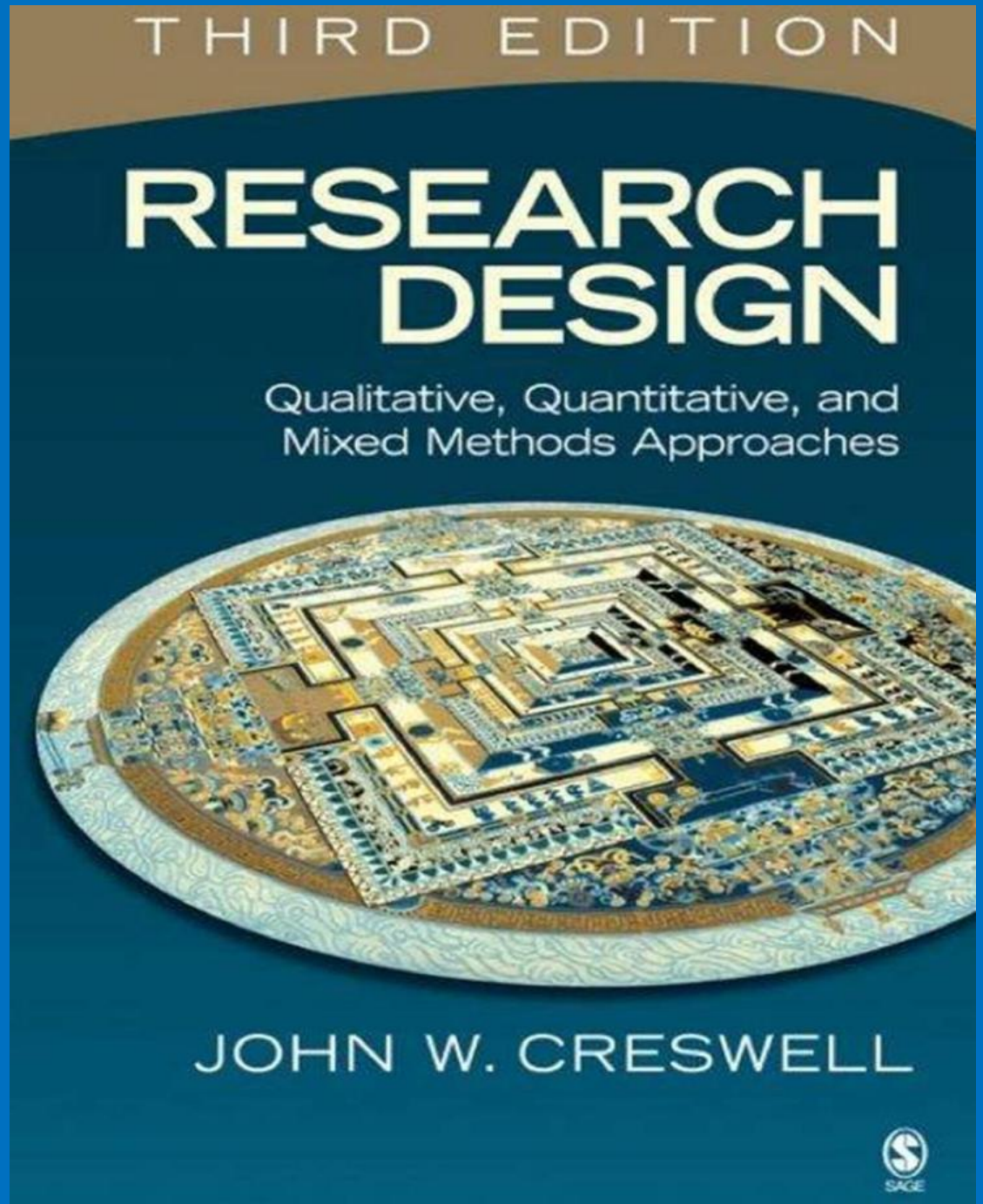
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Recommended Readings



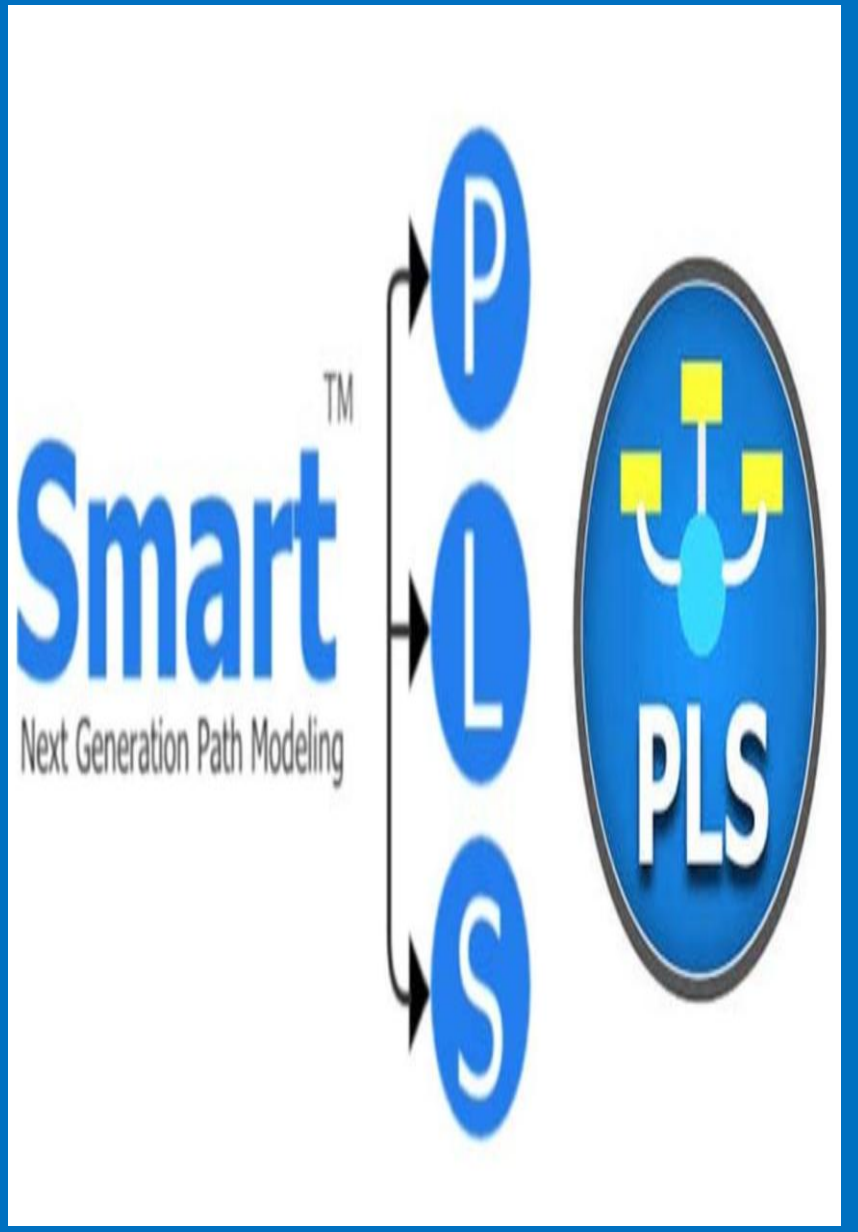


Recommended Readings

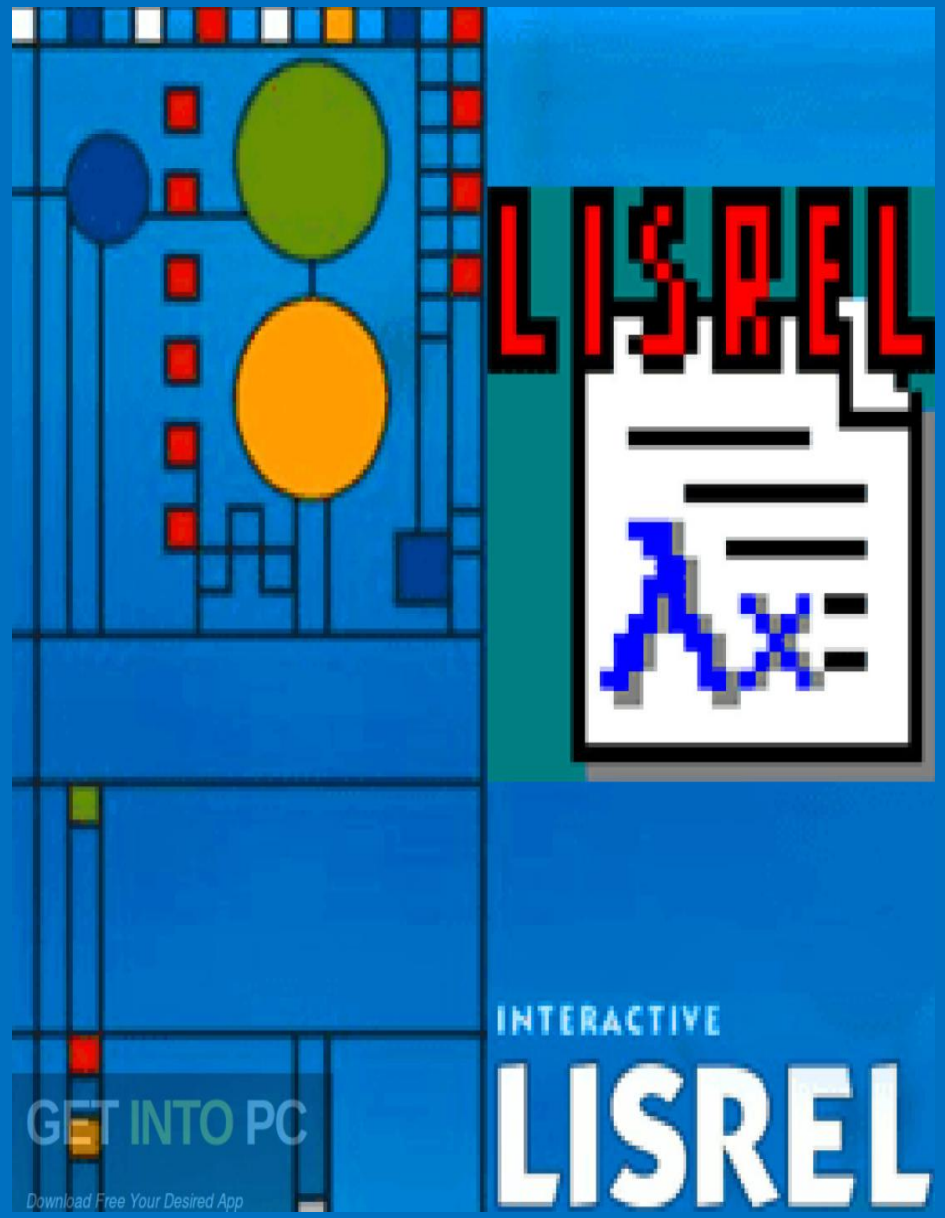


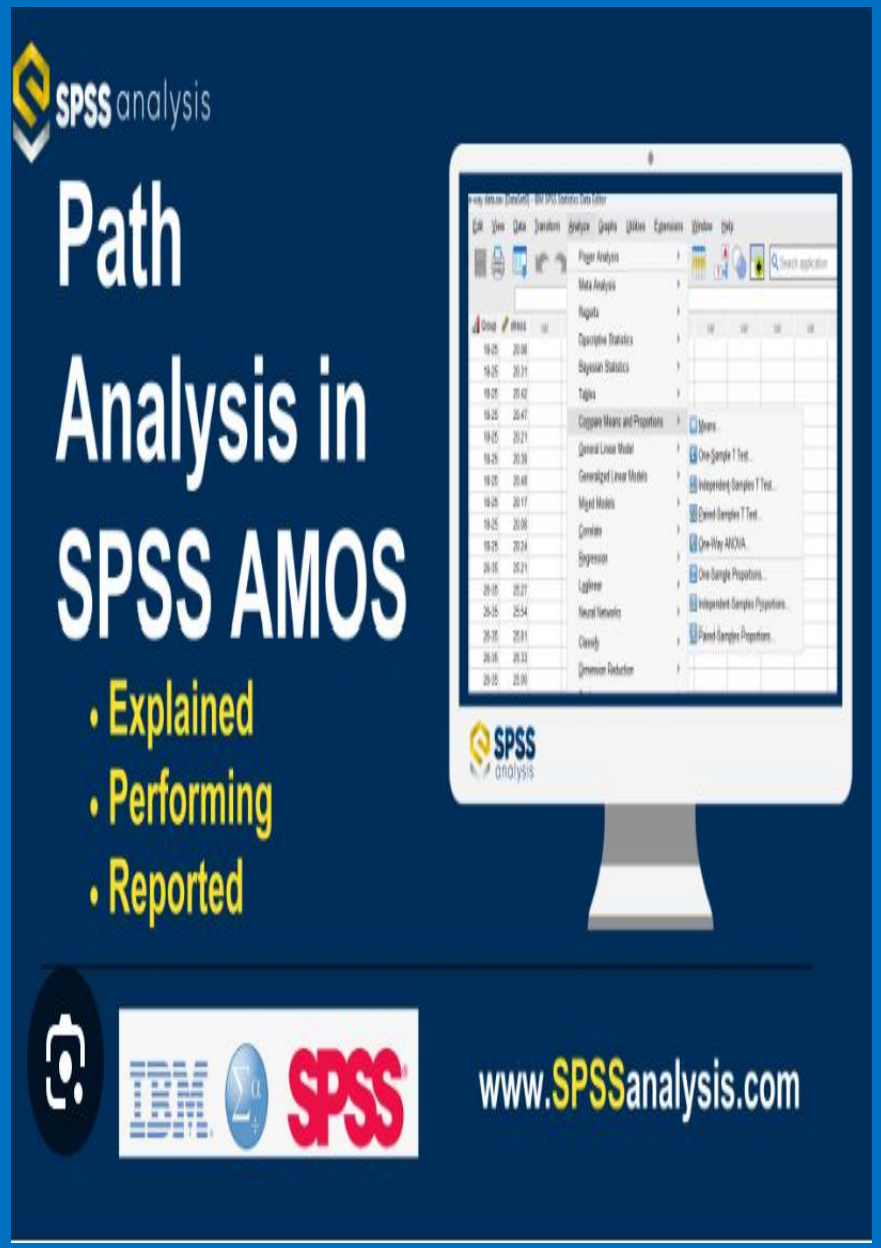
Research Tools



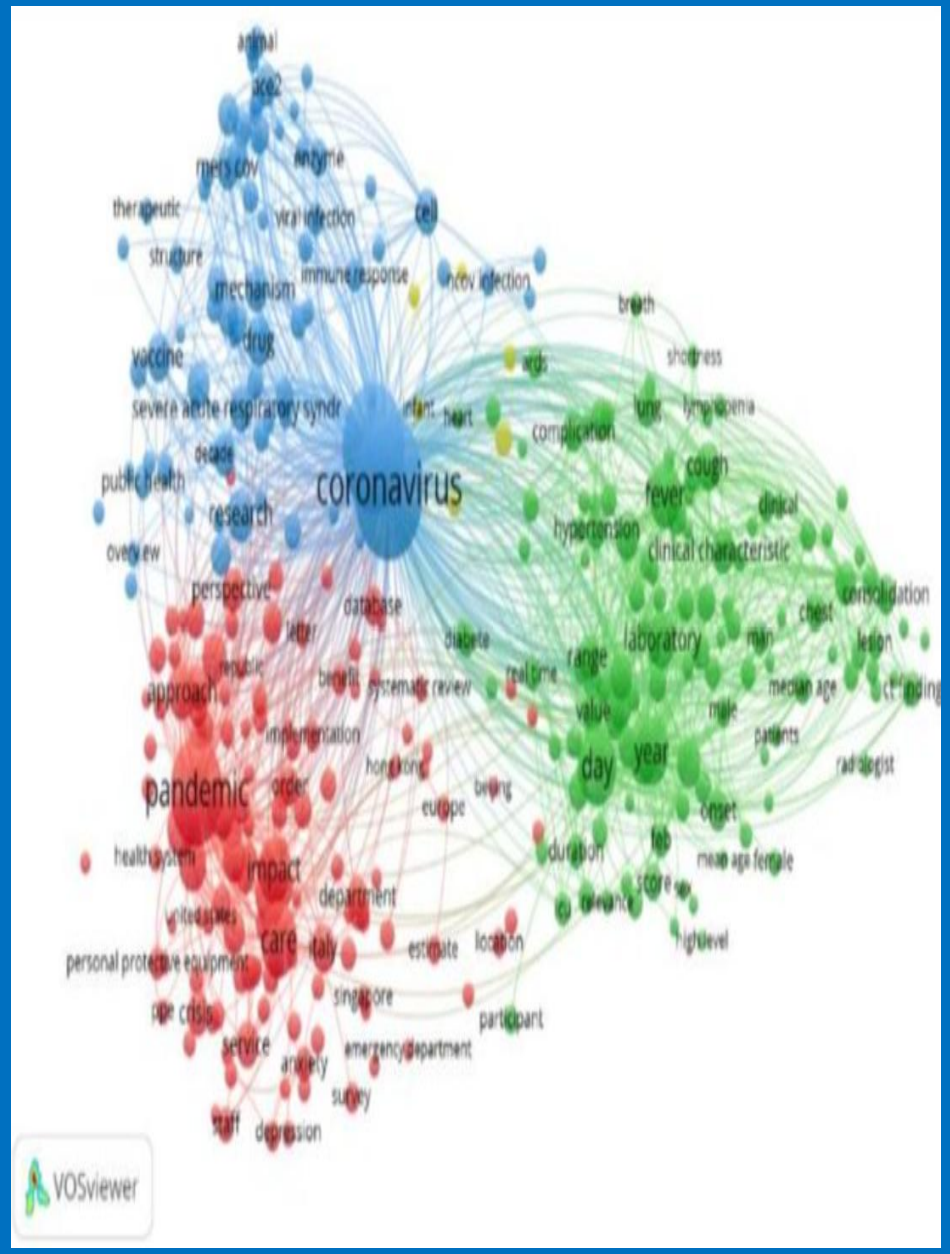


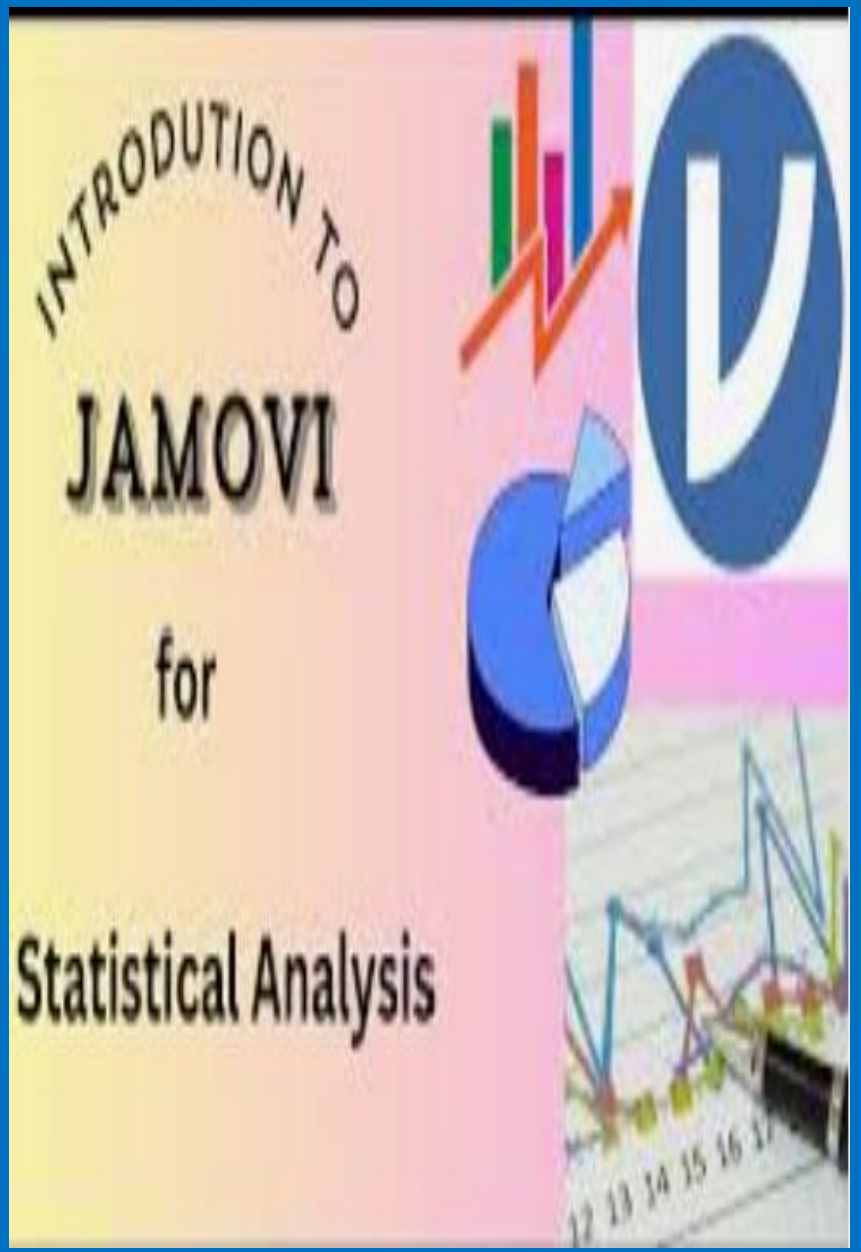
Research Tools





Research Tools





Research Tools





What is Research; Purpose of Research; Research Project Conceptualization; Choice of Methods

What is Research?

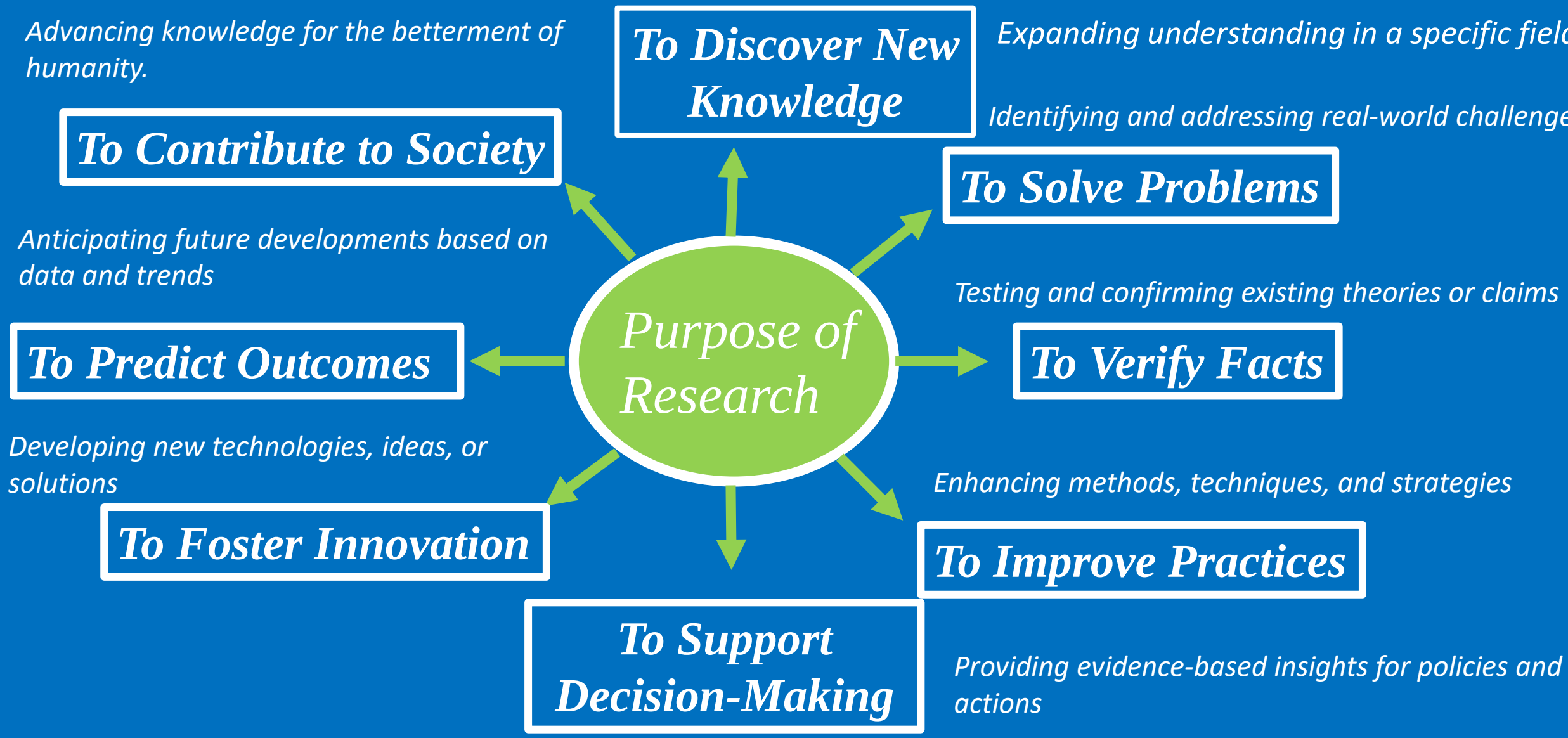
- ✓ *A process of gathering data or information to solve a particular or specific problem in the scientific manner. (Manuel and Medel (1976))*
- ✓ *A systematic study or investigation of something for the purpose of answering questions posed by the researcher. (Sanchez 1980)*
- ✓ *Careful, critical, disciplined inquiry, varying in technique and method according to the nature and condition of problem identified, directed toward clarification or resolution (or both) of a problem.*

What is Research?

Thus,

- ✓ *Research is a systematic investigation conducted to generate new knowledge, validate existing knowledge, or solve specific problems through critical analysis and experimentation.S*

What is the purpose of Research?



Types of Research

- Descriptive Research
- Analytical Research
- Applied Research
- Fundamental Research
- Quantitative Research
- Qualitative Research
- Conceptual Research
- Empirical Research
- One time Research
- Longitudinal Research
- Clinical or Diagnostic Research

Descriptive & Analytical Research

Descriptive Research

- Descriptive means to describe the problem under research.
- Descriptive research is about description of the state of affairs as it exists at present, situation or phenomena.
- Descriptive research focuses on "What?", No Hypothesis is needed
- Examples: frequency of shopping, which flavor of product people like the most.

Analytical Research

- Analytical research is about critically analyzing the facts, data & information already available.
- Analytical research focuses on "How, Why?", Hypothesis is needed
- For example: why such flavor of product people like the most, which age group like most and like less.

Applied & Fundamental Research

Applied (Action) Research

- Applied research is about finding a solution of a problem or issue affecting the group, society or organizations.
- Examples: How to reduce poverty in cities? How to minimize crimes?

Fundamental (Basic/Pure) Research

- Fundamental research is to increase knowledge or development of theory or model.
- Gathering knowledge for the sake of knowledge.
- No solution to a practical problem.
- Example: Is performance is always associated with motivation?
- Academics research (management & social studies)

Qualitative & Quantitative Research

Qualitative Research

- Qualitative research deals with collecting & analyzing non-numerical (subjective) data to understand concepts, opinions or experiences etc.
- Qualitative research is used to understand human behavior, attitude, motivation, satisfaction etc.
- Example: Whether employees are satisfied with current reward system of a company.

Quantitative Research

- Quantitative research is the type of research that is used to collect & analyze numerical data or facts, and find results by applying statistical and mathematical techniques.
- Example: How much time is decreased in product life cycle after implementing new technology?

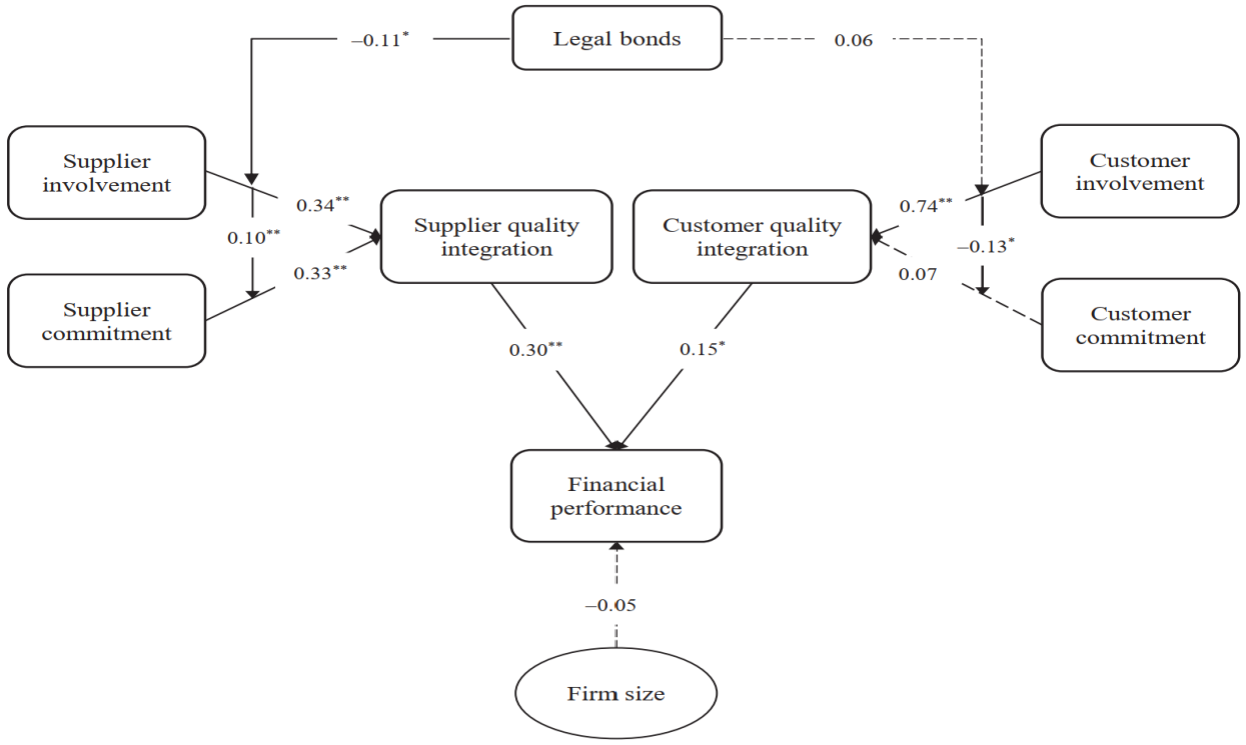
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QUALITY PAPER

The effect of relational investments on supply chain quality integration: protection or constraint of legal bonds?

Faheem Akhtar
School of Management, Zhejiang University, Hangzhou, China
Qianwen Wang
Research Center for Smarter Supply Chain, Soochow University, Suzhou, China, and
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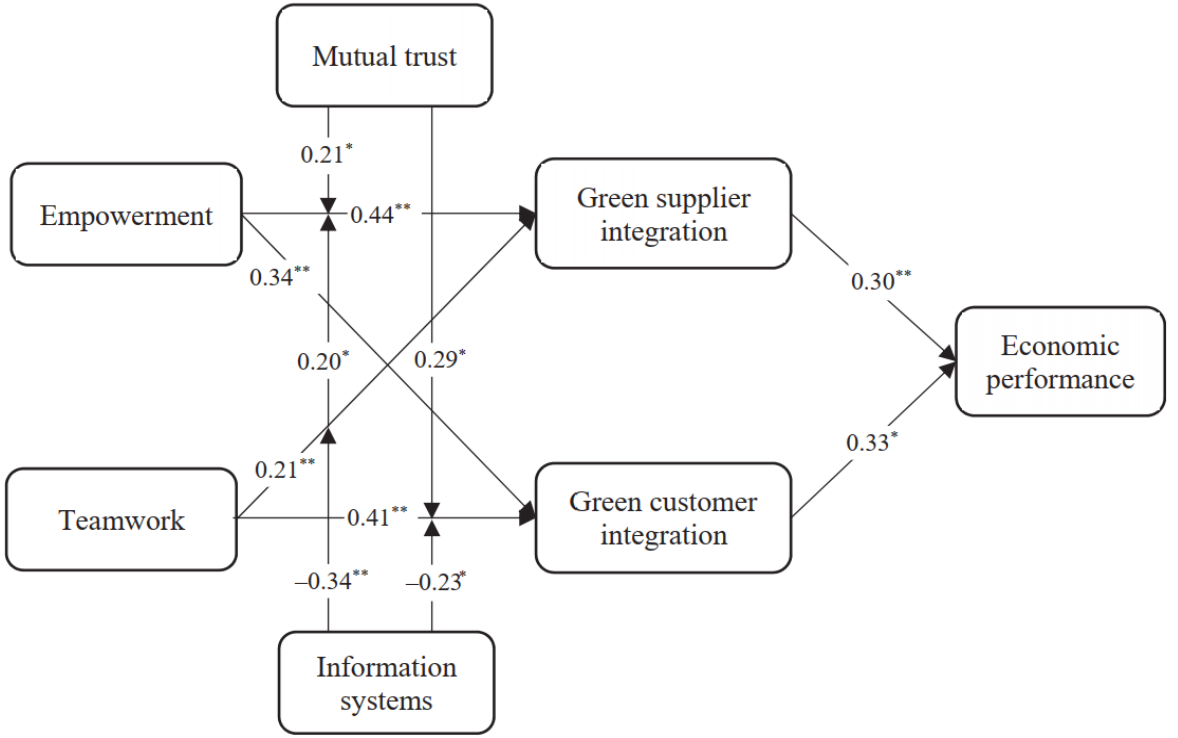


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The effect of human resource strategy on green supply chain integration: the moderating role of information systems and mutual trust

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Qianwen Wang
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Embracing green supply chain collaboration through technologies: the bridging role of advanced manufacturing technology

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Qianwen Wang
Research Center for Smarter Supply Chain, Soochow University, Suzhou, China

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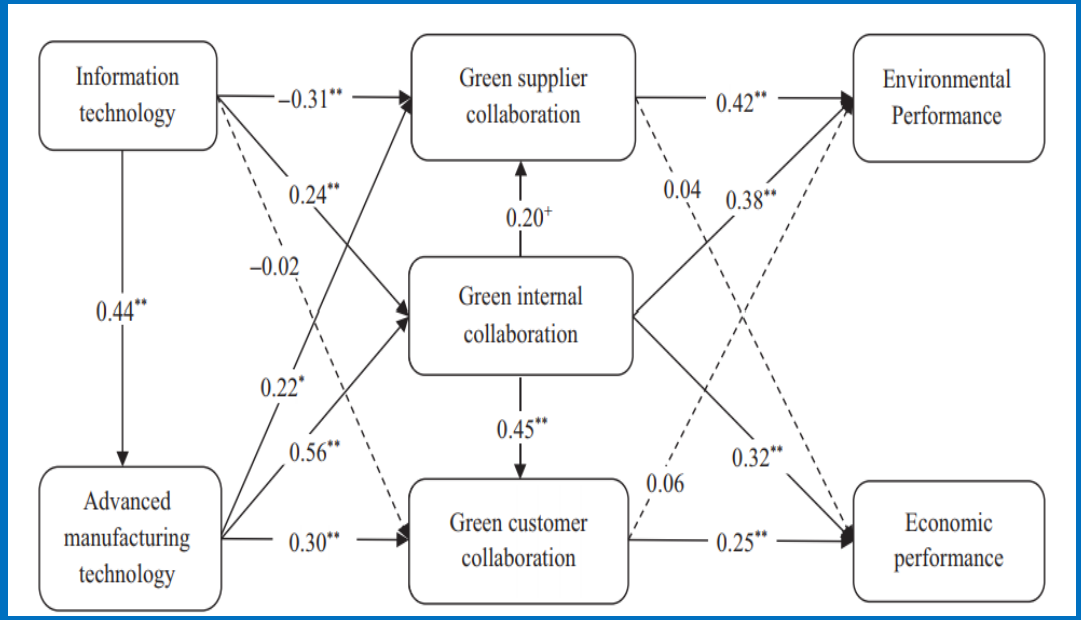


Table 4 Correlation matrix

Constructs	Mean	SD	1	2	3	4	5	6	7
1. IT	3.62	1.605	*0.778*						
2. AMT	4.75	1.506	0.493**	*0.812*					
3. Green supplier collaboration	4.50	1.390	0.462**	0.445**	*0.721*				
4. Green internal collaboration	4.74	1.441	0.475**	0.446**	0.678**	*0.773*			
5. Green customer collaboration	4.35	1.508	0.324**	0.230**	0.585**	0.443**	*0.872*		
6. Environmental performance	5.51	1.022	0.210**	0.147*	0.202**	0.205**	0.171*	*0.852*	
7. Economic performance	4.49	1.262	0.242**	0.124	0.291**	0.261**	0.179**	0.420**	*0.870*

Notes: $^{+}p < 0.10$; $^{*}p < 0.05$; $^{**}p < 0.01$. The diagonal is the square root of the AVE value of each construct
Source: Derived from the statistical analysis of this study

Conceptual & Empirical Research

Conceptual Research

- Philosophers & thinkers used this research to develop new concepts or reinterpret the existing ones.
- Conceptual research is the type of research in which research is conducted by observing and analyzing already available information on a topic under study.
- *No practical experiments is conducted.*

Empirical Research

- *Experiments & observations based research*.
- It is based on observed & measured phenomena and derives conclusion & knowledge from actual experience rather than from theory or belief.
- Conclusions are derived only from concrete & verifiable evidence.

One time & Longitudinal Research

One Time Research

- As the name shows it is the type of research which is carried out over a single time period.
- For example: Population Researches

Longitudinal Research

- A type of research which involves study of sample at more than one point in time.
- For example: market trends, customer satisfaction, employees motivation, brand awareness etc.

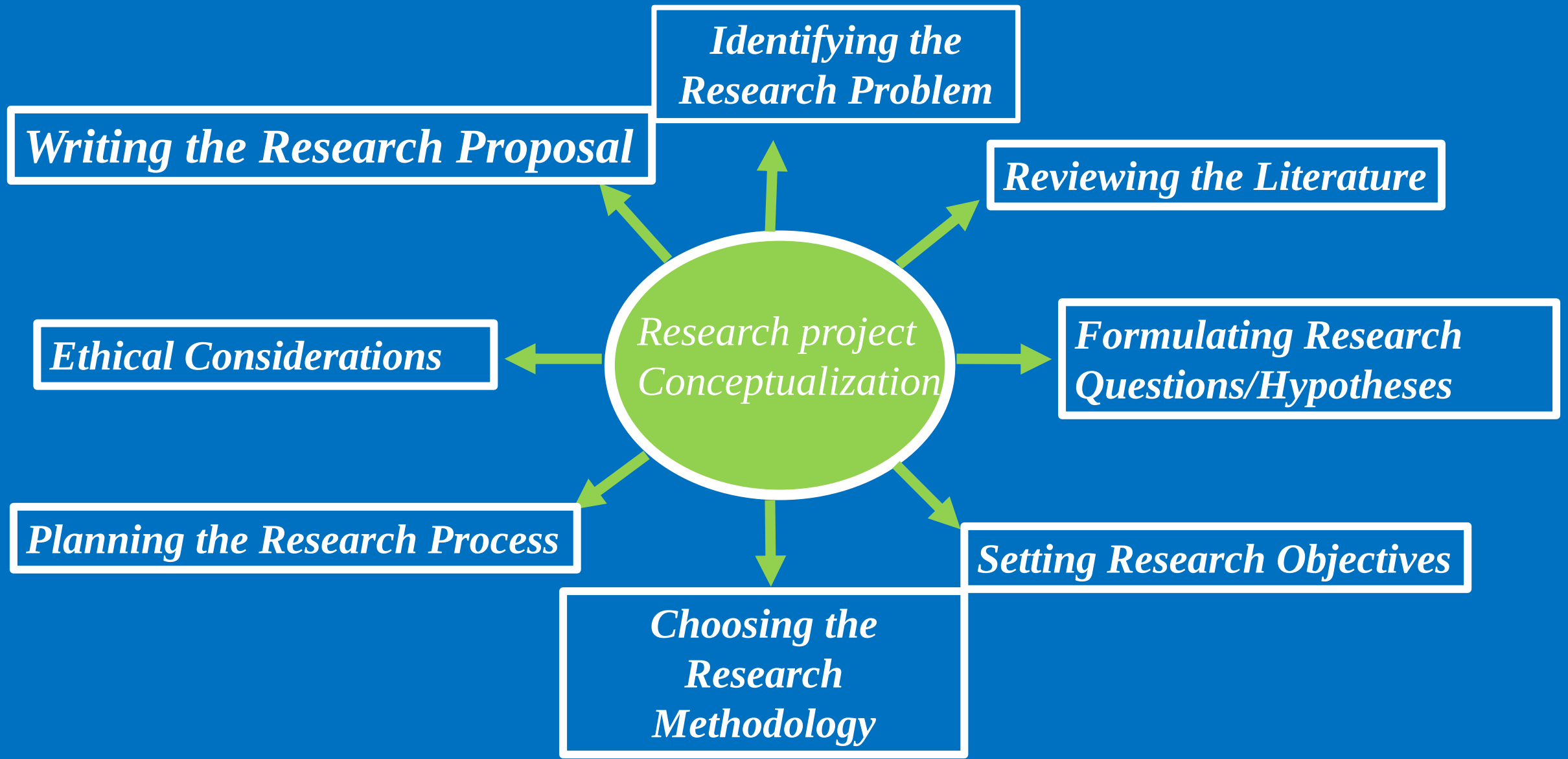
Clinical or Diagnostic Research

- It is the type of research which is carried out over a humans.
- Clinical research improves disease knowledge, treatment procedures, diagnostic techniques etc.
- Example: treatment of cancer, HIV aids etc.

Research Project Conceptualization

- ✓ *Research Project Conceptualization is the initial phase of a research project where the foundation for the study is laid out.*
- ✓ *This phase involves identifying the central problem or topic, formulating research questions, defining objectives, and setting up a framework for the study.*
- ✓ *Essentially, it's the process of turning an idea or problem into a concrete research project.*

Research Project Conceptualization



THANKS